

Sina Parsnejad Publications

Journal publications:

- S. Parsnejad and A. J. Mason, "Etacteme Machine-to-Human Communication: A Linguistically-Inspired Experimentally-Derived Electrotactile Icon Generation Scheme *IEEE Transactions on Biomedical Engineering (TBME)* [in review], 2026.
- S. Parsnejad, E. Ashoori, S. Davila-Montero, and A. J. Mason, "Distinguishable Multi-layer Frequency-modulated Waveforms for Peripheral Nerve Communication in Human Augmentation Applications," *IEEE Trans. Haptics*, pp. 1–12, 2026, doi: 10.1109/TOH.2026.3665317.
- S. Parsnejad, J. W. Brascamp, G. Pelled and A. J. Mason, "A review of electrotactile stimulation for machine-to-human communication," in *IEEE Transactions on Biomedical Engineering*, doi: 10.1109/TBME.2026.3661416.
- Alanazi, M.O.; Patano, A.; Bente, G.; Mason, A.; Goldstein, D.; Parsnejad, S.; Wyatt, G.; Lehto, R. Nature-Based Virtual Reality Feasibility and Acceptability Pilot for Caregiver Respite. *Curr. Oncol.* **2023**, *30*, 5995-6005. <https://doi.org/10.3390/curroncol30070448>
- Yin, Heyu, et al. "Size-fractionated electrochemical quantification for compact monitoring of fine particulate matter." *Microchemical Journal* 168 (2021): 106386.
- H. Li, S. Parsnejad, E. Ashoori, C. Thompson, E. K. Purcell, and A. J. Mason, "Ultracompact Microwatt CMOS Current Readout with Picoampere Noise and Kilohertz Bandwidth for Biosensor Arrays," *IEEE Trans. Biomed. Circuits Syst.*, vol. 12, no. 1, 2018, doi: 10.1109/TBCAS.2017.2752742.

Book chapters:

- Ashoori, Ehsan, Sina Parsnejad, and Andrew J. Mason. "Neurochemical Sensing." *Handbook of Neuroengineering*. Singapore: Springer Nature Singapore, 2023. 591-621.
- Parsnejad, Sina. *Enabling Real-Time Communication for Human Augmentation Systems via Unobtrusive High Bandwidth Machine to Human Electrotactile Peripheral Nerve Stimulation*. Michigan State University, 2022.
- S. Parsnejad and A. J. Mason, CMOS nano-pore technology. 2017. doi: 10.1007/978-3-319-67723-1_1.
- Parsnejad, Sina. Power-Area Efficient Rapid-Response CMOS Frontend for High-Throughput Ion-Channel Sensor Array Microsystems. Michigan State University, 2017.

Conferences

- Dávila-Montero, Sylmarie, et al. "Design of a multi-sensor framework for the real-time monitoring of social interactions." *2022 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, 2022.
- Parsnejad, Sina, et al. "Investigating Distinct intensity levels in electrotactile machine-to-human communication." *2022 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, 2022.
- Parsnejad, Sina, Ehsan Ashoori, and Andrew J. Mason. "Multi-electrode electrotactile stimulation to combat skin dependency in machine-to-human feedback." *2021 IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE, 2021.
- Ashoori, Ehsan, et al. "A Low-Cost Liquid-Based Capacitive Sensor for PM 2.5 Monitoring." *2021 IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE, 2021.
- Dávila-Montero, Sylmarie, Sina Parsnejad, and Andrew J. Mason. "Exploring the relationship between speech and skin conductance for real-time arousal monitoring." *2020 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, 2020.
- Use of high-frequency pulses to generate unique electrotactile sensations for real-time feedback in wearable sensory systems
- Dávila-Montero, Sylmarie, et al. "Improving object localization resolution using a wearable ultrasonic sensor array." *2020 IEEE 63rd International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE, 2020.
- Parsnejad, Sina, Sylmarie Dávila-Montero, and Andrew J. Mason. "Evaluating distinct sensations by modulating high-frequency electrotactile pulses for unobtrusive communication with iot devices." *2020 IEEE 6th World Forum on Internet of Things (WF-IoT)*. IEEE, 2020.
- Yin, Heyu, et al. "Microfabricated capacitive electrodes for high channel count ECoG recording." *2019 9th International IEEE/EMBS Conference on Neural Engineering (NER)*. IEEE, 2019.
- S. Parsnejad, Y. Gtat, R. Aridi, J. W. Brascamp, and A. J. Mason, "Inciting high fidelity tactile sensations using a single electrotactile electrode pair," in *International IEEE/EMBS Conference on Neural Engineering, NER*, 2019. doi: 10.1109/NER.2019.8717131.
- H. Yin, E. Ashoori, S. Parsnejad, J. W. Salatino, E. K. Purcell, and A. J. Mason, "Microfabricated Capacitive Electrodes for High Channel Count ECoG Recording," in *International IEEE/EMBS Conference on Neural Engineering, NER*, 2019. doi: 10.1109/NER.2019.8717074.
- Y. Gtat, S. Parsnejad, S. Schrand, H. L. Taylor, and A. J. Mason, "Absolute Detection Threshold of Vibrotactile Stimulation Pulse Width and Inter-Pulse Gap," in *International IEEE/EMBS Conference on Neural Engineering, NER*, 2019. doi: 10.1109/NER.2019.8716900.
- Y. Gtat, S. Parsnejad, and A. J. Mason, "Platform for autonomous sensor characterization and generation of provenance-aware datasets," in *Midwest Symposium on Circuits and Systems*, 2019. doi: 10.1109/MWSCAS.2018.8623989.
- S. Parsnejad, Y. Gtat, T.-Y. Lin, X. Liu, P. B. Lillehoj, and A. J. Mason, "Self-ranging thumb-sized multichannel electrochemical instrument for global wearable point-of-care sensing," in *Midwest Symposium on Circuits and Systems*, 2019. doi: 10.1109/MWSCAS.2018.8623887.
- Ashoori, Ehsan, et al. "ECoG electrode array with embedded coupling capacitors for area efficient neural recording." *2018 IEEE Biomedical Circuits and Systems Conference (BioCAS)*. IEEE, 2018.
- Y. Gtat, S. Parsnejad, and A. J. Mason, "Live demonstration: Automated data acquisition and digital curation platform for enhancing research precision, productivity and reproducibility," in *Proceedings - IEEE International Symposium on Circuits and Systems*, 2017. doi: 10.1109/ISCAS.2017.8050711.
- S. Parsnejad, Y. Hu, H. Wan, E. Ashoori, and A. J. Mason, "Wide dynamic range multi-channel electrochemical instrument for in-field measurements," in *Proceedings of IEEE Sensors*, 2017. doi: 10.1109/ICSENS.2016.7808670.
- S. Parsnejad, H. Li, and A. J. Mason, "Compact CMOS amperometric readout for nanopore arrays in high throughput lab-on-CMOS," in *Proceedings - IEEE International Symposium on Circuits and Systems*, 2016. doi: 10.1109/ISCAS.2016.7539187.
- Parsnejad, Sina, et al. "Self-ranging thumb-sized multichannel electrochemical instrument for global wearable point-of-care sensing." *2018 IEEE 61st International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE, 2018.
- H. Li, S. Parsnejad, and A. J. Mason, "Single ion channel CMOS electrochemical instrument for high throughput recording arrays," in *Midwest Symposium on Circuits and Systems*, 2015. doi: 10.1109/MWSCAS.2015.7282111.
- S. Parsnejad, M. Akcakaya, and G. Dundar, "A low power second order current mode continuous time sigma delta ADC with 98 dB SNDR," in *Conference Proceedings - 10th Conference on Ph. D. Research in Microelectronics and Electronics, PRIME 2014*, 2014.